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Math

Extra

Series solution:

$$xy'' + y' + xy = 0$$

if $y = e^x, \ln x, \cos x$ etc.

or, $y = a + bx + cx^2 + dx^3 + \dots$

$$x^2 y'' + xy' + x^2 y = 0$$

$$x \frac{d}{dx} \left(x \frac{dy}{dx} \right) + x^2 y = 0$$

$$y = a_0 + a_1 x + a_2 x^2 + \dots$$

$$y = \sum a_n x^n$$

$$\frac{dy}{dx} = \sum a_n n x^{n-1}$$

$$x \frac{dy}{dx} = \sum a_n n x^n$$

$$\sum n^2 a_n x^n + \sum a_n x^{n+2} = 0$$

$$\sum_{n=0}^{\infty} n^2 a_n x^n + a_0 x^2 + a_1 x^3$$

Recurrence Relation

$$n+2 = n$$

$$n = n-2$$

$$n = 2$$

$$\sum_{n=0}^{\infty} n^2 a_n x^n + \sum_{n=2}^{\infty} a_{n-2} x^n$$

$$\sum_{n=0}^{\infty} n^2 a_n x^n$$

$$+ \sum_{n=2}^{\infty} n^2 a_n x^n + \sum_{n=2}^{\infty} a_{n-2} x^n$$

$$n^2 a_n + a_{n-2} = 0$$

$$a_n = - \frac{a_{n-2}}{n^2}$$

Recurrence formula

Initial conditions

$$a_0 = 1$$

$$a_2 = - \frac{a_0}{2^2}$$

$$a_4 = 0$$

$$a_6 = - \frac{a_2}{4^2} = \frac{a_0}{2^2 \cdot 4^2}$$

$$a_8 = 0$$

$$a_{10} = \dots$$

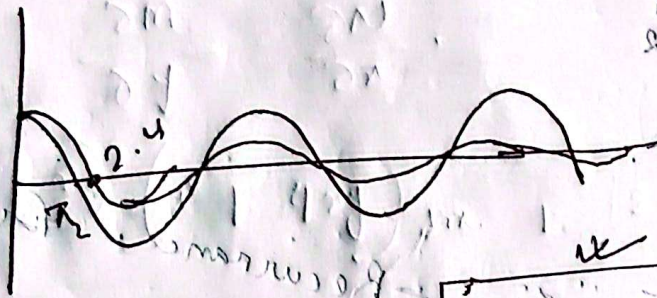
zeroth Bessel's solution

$$y = a_0 \left(1 - \frac{x^2}{2^2} + \frac{x^4}{2^2 \cdot 4^2} - \dots \right)$$

→ solution of power series.

Two arbitrary constant a_0 and a_1 are given.

$$a_0 \cos x = a_0 \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \right)$$



$$y = a_0 \cos x + a_1 \sin x$$

② $y dx + x dy = 0$ → $y \frac{dx}{x} = - \frac{dy}{y}$

Integrating factor my both side → $\frac{dx}{x} = - \frac{dy}{y}$

$$\frac{dx}{x} + \frac{dy}{y} = 0$$

→ Integrate both side,

$$\ln x + \ln y = 0$$

Variable separation

or ans?

① $f(x) \cdot g(y) dx + f(x) \cdot f(y) dy = 0$

$x^2 \cos y dx + (x^2 + 1) \sin y dy = 0$

'x' function
'y' function

$\frac{x^2}{x^2 + 1} dx + \frac{\sin y}{\cos y} dy = 0$

or ans?

② $(x^2 + y) dx + y dy = 0$

$\frac{x^2 dx}{\text{Integrate}} + \frac{y dy}{\text{Integrate}} + \frac{y dy}{\text{Integrate}}$

Exact solution

Exact

first order differ

$\frac{dy}{dx} = f(x, y)$

$\frac{dy}{dx} = \frac{x^2}{y+2}$

$(y+2) dx + x^2 dy = 0$

$M dx + N dy = 0 = d(\phi)$
Total differentiation form
represent as to ans
Exact soln.

where $q(p) = \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy$

$$\frac{\partial \phi}{\partial x} = m \quad \left| \quad \frac{\partial \phi}{\partial y} = N \right.$$

$$\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial m}{\partial y} \quad \left| \quad \frac{\partial^2 \phi}{\partial y \partial x} = \frac{\partial N}{\partial x} \right.$$

Exact,

$$\frac{\partial m}{\partial y} = \frac{\partial N}{\partial x}$$

(*) $(x^3 + 4xy) dx + (2x^2 + y^2) dy = 0$

Homogeneous 2: (m), non homogeneous.

$$(x^3 + 4x^{\frac{2}{2}}y) dx + (2x^2 + y^2) dy = 0$$

variable 2: power same in both terms homogeneous

homogeneous $\rightarrow x^2 dx + xy dy + y^2 dy = 0$

$$(p) = \begin{bmatrix} 0 = p^2 + 1 + q^2 \end{bmatrix}$$

$$\frac{d\phi}{dx} = \phi \left(\frac{y}{x} \right) \rightarrow dx \text{ (but } y \text{ and } dy \text{ are not common)}$$

$$x^3 dx + \frac{4xy dx + 2x^2 dy}{\text{multiplying '1' to } y} + y^2 dy = 0$$

$$\int x dx = \int d\left(\frac{x^2}{2}\right) \text{ same.}$$

$$\hookrightarrow \int \frac{1}{2} \cdot 2x dx = \int x dx$$

$$\Rightarrow x^3 dx + d(2x^2 y) + d\left(\frac{y^3}{3}\right) = 0$$

$$d\left(\frac{x^4}{4}\right) + d(2x^2 y) + d\left(\frac{y^3}{3}\right) = 0$$

$$\Rightarrow d\left(\frac{x^4}{4} + 2x^2 y + \frac{y^3}{3}\right) = 0$$

∴ $d(\phi)$ zero, then ϕ is constant.

∴ ϕ (or) integrate ϕ is constant.

$$\therefore \frac{x^4}{4} + 2x^2 y + \frac{y^3}{3} = C$$

↑
solution.

∴ $\frac{x^4}{4}$ term integrate $\frac{1}{4} \cdot 4x^3 = x^4$ grouping

Linear $\rightarrow$ Exact, 26.41. In $\left(\frac{1}{x}\right)$ If we put,

$$0 = \frac{1}{x} b + \frac{1}{x^2} b + \dots + \frac{1}{x^n} b$$

Linear.

if power '1' is not

$$\left[\frac{1}{x} b \right] = \frac{1}{x} b$$

$$0 = \left(\frac{1}{x} b \right) + \left(\frac{1}{x^2} b \right) + \dots + \frac{1}{x^n} b$$

$$0 = \left(\frac{1}{x} b \right) + \left(\frac{1}{x^2} b \right) + \left(\frac{1}{x^3} b \right)$$

$$0 = \left(\frac{1}{x} b \right) + \left(\frac{1}{x^2} b \right) + \left(\frac{1}{x^3} b \right)$$

if we consider (b) as

$$b = \frac{1}{x} + \frac{1}{x^2} + \frac{1}{x^3}$$

if we consider (b) as